

Rachel Bosco

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PROFESSIONAL SUMMARY

Final-year B.Tech student specializing in AIML with a CGPA of 8.4/10 and 6 months of internship experience in AI/ML development. Hands-on experience across 8+ machine learning and computer vision projects, including deployed AI applications. Proficient in Python, PyTorch, TensorFlow, Scikit-learn, and YOLO. Skilled in predictive modeling, NLP, and developing AI-driven solutions for real-world applications.

EDUCATION

B.Tech in Computer Science (AI&ML) – CGPA: 8.4/10

Garden City University

2022 - 2026

Pre-University (PCMC) – 76%

Jyoti Nivas Pre-University College

2020 - 2022

High School (SSLC Board) – 89%

St. Euphrasia's Girls High School

2017 - 2020

Primary School, St. Agnes' Girls Higher Primary School

2007 - 2017

SKILLS

Programming Languages: C++, Python, HTML5, CSS3, SQL

Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, MediaPipe, Tesseract OCR

Web & Deployment: Flask, Streamlit, REST APIs, Git/GitHub

Specializations: Machine Learning, Computer Vision, Deep Learning, Natural Language Processing, Model Explainability (SHAP)

EXPERIENCE

Data Science & ML Intern

Jan 2026 - Apr 2026

Zaalima Development Pvt. Ltd., Bengaluru

- Worked on AI-powered predictive maintenance solutions using Machine Learning, LightGBM, and time-series feature engineering for industrial machine failure prediction from IoT sensor data.
- Contributed to data preprocessing, SHAP-based model explainability, Flask API deployment, and interactive dashboard integration for real-time risk analysis and monitoring workflows.
- Developed AI-driven industrial quality inspection solutions using Computer Vision, Deep Learning, and PyTorch for automated defect detection applications.
- Assisted in model training, real-time inference pipeline development, and Flask-based deployment workflows for scalable industrial inspection systems.

AIML Intern

Oct 2025 - Jan 2026

Internz Learn Pvt Ltd, Bengaluru

- Developed AI-powered analytics systems including sales forecasting and resume screening using Python, NLP, and machine learning techniques.
- Built forecasting and candidate-matching models using Facebook Prophet, TF-IDF embeddings, and cosine similarity for predictive analysis and ranking.
- Performed data preprocessing, feature engineering, model evaluation, and data visualization to generate actionable business insights.

PROJECTS

VisionSpec QC – PCB Quality Inspection System

- Developed a deep learning-based PCB defect detection system using MobileNetV2 transfer learning for automated PASS/DEFECT classification, achieving 96.68% test accuracy with optimized real-time inference performance.
- Integrated Grad-CAM explainability and deployed a Flask-based inference dashboard for scalable industrial quality inspection workflows.

SparkScale Churn — Distributed Telecom Churn Prediction Platform

- Engineered a distributed telecom churn prediction platform using Apache Spark and PySpark MLlib for scalable feature engineering and predictive analytics, achieving an AUC-ROC score of 0.84.
- Architected and deployed an interactive analytics dashboard enabling real-time churn inference, customer risk stratification, batch prediction workflows, and model performance monitoring.

FabricScan AI – Fabric Identifier System

- Trained an Inception-based deep learning classifier across 8 fabric categories achieving 95.8% peak accuracy for wool; final-year capstone project.
- Deployed a Streamlit web app with voice assistant, 3D texture visualization, and care/recommendation engine; stored all user sessions in PostgreSQL for analytics.

Real-Time Moving Object Detection using OpenCV

- Developed a real-time moving object detection system using OpenCV and Python by implementing frame differencing, contour detection, thresholding, and Gaussian blur techniques for motion analysis.
- Built image preprocessing modules for resizing, smoothing, and real-time video stream processing using webcam-based computer vision pipelines.

Age & Gender Detection

- Developed a real-time age and gender detection system using OpenCV DNN and pre-trained Caffe/TensorFlow models for live webcam-based inference.
- Implemented face detection, age classification, and gender prediction pipelines with optimized real-time video processing and computer vision techniques.

ACHIEVEMENTS

Winner – Smart India Hackathon 2024

Declared Winner at the event organized by Dr. A.P.J Abdul Kalam School of Engineering, Garden City University.

CERTIFICATIONS

- Mastering Big Data Analytics with PySpark – Infosys Springboard
- Hadoop Architecture and HDFS – Infosys Springboard
- CCNA: Introduction to Networks – Cisco
- Python Essentials – Infosys Springboard
- Cyber Security – Cisco